

1. Administrative & Study Identification

Full Study Title: A PHASE 1, RANDOMIZED, DOUBLE-BLIND, SPONSOR-OPEN, PLACEBO CONTROLLED, DOSE ESCALATING STUDY TO EVALUATE THE SAFETY, TOLERABILITY, PHARMACOKINETICS, AND PHARMACODYNAMICS OF SINGLE AND MULTIPLE SUBCUTANEOUS DOSES OF PF-07314470 IN HEALTHY PARTICIPANTS **Short Title/Acronym:** A Study to Learn How Different Amounts of the Study Medicine Called PF-07314470 Are Tolerated and Act in the Body in Healthy Adults **Protocol Number:** C5271001 **NCT Number:** NCT06821750 **Sponsor Name:** Pfizer **Investigational Product:** PF-07314470 (Bispecific antibody targeting IL-27RA and gp130 / IL-27 and IL-6 inhibitor) **Phase of Development:** Phase 1 **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, pharmacokinetics (PK), and pharmacodynamics (PD) of single and multiple subcutaneous doses of PF-07314470 in healthy participants. **Secondary Objectives:** To establish the initial human safety profile, understand drug behavior in the body, and evaluate early evidence of target engagement.

2.2 Background & Rationale PF-07314470 is an investigational bispecific antibody designed to simultaneously modulate two key cytokine pathways (IL-27 and IL-6) implicated in inflammatory and autoimmune disorders (such as atopic dermatitis and inflammatory bowel diseases). This first-in-human foundational study is required to establish the safety, tolerability, and PK/PD profile in healthy volunteers before evaluating its efficacy in patients with specific immunopathological medical conditions. *(Note: This trial was terminated in August 2025 for sponsor business reasons; no safety concerns were identified).*

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind, Sponsor-open **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 23 (Actual enrolled prior to trial termination) **Number of Sites:** Clinical trial units in the United States **Estimated Study Start:** 11-Feb-2025 (Actual) **Estimated Primary Completion:** 18-Aug-2025 (Actual/Terminated)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy males aged 18 to 45 years and healthy females aged 18 to 55 years. 2. Body Mass Index (BMI) of 16-32 kg/m², and a total body weight greater than 50 kg (110 lb). 3. For Japanese participants only, a total body weight greater than 45 kg and having 4 biological grandparents who were born in Japan.

4.2 Exclusion Criteria 1. Significant medical issues or a history of autoimmune/inflammatory diseases. 2. Excessive use of tobacco or alcohol. 3. Conditions or factors that could impact drug absorption, safety evaluation, or compliance with study procedures.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Dose-escalation design. Participants receive PF-07314470 weekly at 1 of 4 dose levels for 4 doses, or every 2 weeks at 1 dose level for 3 doses (or matching placebo). **Route of Administration:** Subcutaneous (SC) Injection **Duration of Treatment:** Up to 3 to 4 weeks of dosing, followed by a safety follow-up period.

5.2 Concomitant & Prohibited Therapy Permitted: Standard over-the-counter medications strictly approved by the investigator for healthy volunteers (e.g., occasional paracetamol). **Prohibited:** Use of other investigational drugs, current immunosuppressive therapies, biologics, or concomitant medications that could confound the assessment of the study drug.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -1	Informed Consent, Medical History, BMI evaluation, Eligibility Labs
Baseline	Day 1	Randomization, First SC Dose, Vital Signs, ECG
Interim	Weekly/Bi-weekly	Subsequent Doses, Safety Labs, PK/PD blood draws, AE Monitoring
End of Study	Follow-up	Final Vital Signs, Exit Interview, Safety Labs, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence and severity of Treatment-Emergent Adverse Events (TEAEs) and Serious Adverse Events (SAEs) to evaluate overall safety and tolerability. **Measurement Method:** Clinical assessments, physical exams, vital signs, ECGs, and central laboratory analyses.

7.2 Secondary & Exploratory Endpoints 1. **Pharmacokinetics (PK):** Maximum observed concentration ($SC_{\max}$) and Area Under the Curve (AUC) of PF-07314470. 2. **Pharmacodynamics (PD):** Biomarker tracking to provide early evidence of target engagement and biological activity for IL-27 and IL-6 pathway modulation.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via participant reporting and clinical site assessments throughout the treatment and follow-up phases. **Serious Adverse Events (SAEs):** Required reporting to the sponsor and Institutional Review Board (IRB) within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal safety review committee responsible for reviewing data from each cohort prior to advancing to the next dose-escalation level.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all randomized participants who received at least one dose of the investigational product). **Sample Size Justification:** Descriptive statistical design standard for Phase 1 dose-escalation studies; not powered for statistical significance, but suitably sized to identify initial safety trends and PK/PD profiles. **Handling of Missing Data:** Standard imputation methods for safety and precise observed data models for PK/PD parameters.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring by the sponsor to ensure protocol adherence and participant safety. **Audit Trail:** Detailed electronic Case Report Form (eCRF) locking and formalized data clarification processes.

11. Study Identifiers & Governance

Primary ID: C5271001 **Registry Number:** NCT06821750 **Sponsor/Lead Organization:** Pfizer **Study Title:** A Study to Learn How Different Amounts of the Study Medicine Called PF-07314470 Are Tolerated and Act in the Body in Healthy Adults **Official Regulatory Title:** A PHASE 1, RANDOMIZED, DOUBLE-BLIND, SPONSOR-OPEN, PLACEBO CONTROLLED, DOSE ESCALATING STUDY TO EVALUATE THE SAFETY, TOLERABILITY, PHARMACOKINETICS, AND PHARMACODYNAMICS OF SINGLE AND MULTIPLE SUBCUTANEOUS DOSES OF PF-07314470 IN HEALTHY PARTICIPANTS

12. Clinical Framework (Phase 1 / Healthy Volunteer Specific)

Primary Condition: Healthy Volunteers **Diagnosis Threshold:** Healthy status confirmed by extensive screening labs, medical history, and BMI 16-32 kg/m2. **Medical Methodology:** Phase 1 Safety, Tolerability, and Dose-Escalation Evaluation **Optimization Target:** Establishing Maximum Tolerated Dose (MTD) and initial human PK/PD parameters.

13. Study Procedures & Methodology

Management Pathway: Confined/outpatient Phase 1 clinical trial unit standardized pathway **Education Protocol (HCP):** Protocol training, precise SC administration instructions, and strict dose-escalation stopping rules **Education Protocol (Patient):** Informed consent detailing first-in-human risks and Phase 1 trial lifestyle restrictions (diet, smoking, alcohol) **Quality Audit Mechanism:** Sponsor medical oversight, strict inclusion/exclusion validation, and IRB periodic safety reviews

14. Observation & Follow-Up Schedule

Total Duration: Approximately 8-12 weeks per participant (including screening, dosing, and follow-up) **Onsite Visit Cadence:** Frequent. Weekly or bi-weekly depending on the dose-escalation cohort assignment **Remote Monitoring Frequency:** N/A (Predominantly reliant on intensive onsite visits or short clinical confinements) **Checklist Review Interval:** Every clinical visit for AE/SAE and concomitant medication tracking

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				